

SAMUEL J. TUSICK

724-601-4365 | samtusick@outlook.com | linkedin.com/in/samuel-tusick/ | samtusick.github.io/PortfolioWebsite/

PROFESSIONAL SUMMARY

Software Engineering student (4.0 in-major GPA) building AI-powered, cloud-native, full-stack applications. Contributor to faculty-supervised research on LLM-integrated tutoring systems and a serverless AWS virtual-patient training platform. Experienced leading development teams and delivering software with faculty, researchers, and cross-functional stakeholders.

EDUCATION

Florida Gulf Coast University — B.S. Software Engineering

Fort Myers, FL | Expected May 2027

- **In-Major GPA:** 4.0 | Dean's List (2023, 2024, 2025)
- **Honors:** Ruth M. and Cleveland L. Campbell Engineering Scholarship Endowed Fund, Kunkle Family Scholarship Endowed Fund
- **Relevant Coursework:** Machine Learning (Spring 2027), AI Agent Studio (Spring 2027), Software Architecture & Design (Spring 2027), Intro to Data Engineering, Software Testing, Requirements Engineering & Analysis, Intro to Cloud Computing, Data Structures & Algorithms, Software Engineering Fundamentals, Assembly Language Programming, Computer Security, Operating Systems, Calculus III, Differential Equations

TECHNICAL SKILLS

- **Languages:** C, C++, Python, JavaScript, SQL
- **Applied AI:** Agentic AI system design, LLM-integrated tutoring systems, prompt-driven application design
- **Cloud:** AWS (API Gateway, Lambda, Aurora, Cognito, IAM)
- **Backend:** Flask, PostgreSQL, Supabase, SQLite
- **Frontend:** React, Vite, Tailwind CSS
- **Dev Tools:** Git, GitHub Actions, Selenium, Jira, pgAdmin, VS Code, CLion

RELEVANT EXPERIENCE

FGCU Dendritic AI & DS Institute — Student Research Assistant

Mar 2025 – Present | On-site

- Designed, developed, and deployed 14+ custom AI tutors actively used within the Engineering Learning Hub.
- Coordinated trainings, workshops, and webinars supporting 500+ participants and 150+ hours of instructional content across the 2026 FGCU Summer AI Academy.
- Evaluated emerging Agentic AI frameworks and recommended technologies for future educational and operational deployments.

Router-Level Security Detection — Subteam Lead & Backend Developer

Feb 2025 – Apr 2025 | On-site

- Led a five-member software engineering team responsible for backend architecture, task coordination, technical decision-making, and milestone delivery on an ongoing Eagle CyberNest Lab research project enhancing a home-network security platform.
- Built Flask backend services integrating three router platforms, using SSH and SQLite to enable secure, real-time monitoring and centralized data collection across multiple network devices.
- Tested and deployed production-ready router-communication scripts, improving system reliability and security.

PROJECTS

MAIVA – AI Virtual Patient Training Platform – React, Vite, Tailwind CSS, AWS (API Gateway, Lambda, Aurora, Cognito, IAM) — In Progress

- Developing an AI-powered virtual patient platform supporting graduate mental health education through realistic LLM-driven patient interactions.
- Contributing to system design and requirements specification, including structured data models for patient personas, reviewed by faculty and clinical supervisors, alongside planned Unreal Engine-based avatar rendering.
- Building the React, Vite, and Tailwind CSS frontend and integrating it with a serverless AWS backend (API Gateway, Lambda, Aurora, Cognito, IAM).

Switch4Good Database System – PostgreSQL, SQL, Data Engineering

- Led development of a PostgreSQL database supporting nonprofit operations across 100+ universities and 1,500+ students.
- Designed a PostgreSQL schema to manage student participation, projects, institutional partnerships, and program outcomes.

NHL Stat Scraper – Python, Flask, Selenium, React, Vite, Render, Netlify

- Built a full-stack analytics application allowing users to search and compare live NHL player statistics.
- Built a Flask backend using Selenium to scrape data from the official league site and expose it through REST APIs.
- Developed a React frontend with Vite enabling users to filter and dynamically view stat categories (e.g., goals, assists) via query parameters.

Pinterest Testing Suites – Java, Selenium

- Led a software testing team using Selenium and Agile testing methodologies to evaluate functionality, usability, and regression scenarios.
- Developed Selenium-based test suites in Java and presented findings on functionality, reliability, and usability.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

EXTRACURRICULARS

Computer Science & Software Engineering Club | Member

August 2024 – Present

Deeptech Club | Member

June 2026 – Present



Portfolio Website